Adaptive Output-Feedback Fault-Tolerant Control for Distributed Optimization of Nonlinear MASs Under Event-Triggered Communication

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***Abstract*—This article investigates the distributed adaptive optimization of multiagent systems with uncertain nonlinearities, utilizing a dynamic event-triggered mechanism. Distinct from existing algorithms that assume healthy operation, the proposed dynamic event-triggered strategy not only saves communication resources but also ensures convergence to a small neighborhood of the global minimizer even in the presence of actuator faults, thereby effectively avoiding the loss of optimality or optimization failure caused by malfunctions. Furthermore, considering that practical systems often possess only measurable outputs and exhibit complex nonlinear characteristics, a state observer is developed to estimate the unmeasured state variables. On this basis, by incorporating the backstepping design technique from the nonlinear fault-tolerant control theory, an adaptive event-triggered output feedback fault-tolerant control strategy is designed to compensate for actuator faults. Meanwhile, a high-order filter is introduced to resolve the incompatibility between the signal derivative discontinuity induced by event-triggering and the differentiability requirements of backstepping design. Thereby, the controller design and stability analysis can proceed within a continuous-time framework. Theoretical analysis demonstrates that the proposed method guarantees that all signals in the resulting closed-loop system are bounded. The validity of the proposed control strategy is further verified through a simulation.**

***Index Terms*—Actuator faults, adaptive backstepping control, distributed optimization, event-triggered communication, fault-tolerant control (FTC), nonlinear multiagent systems (MASs).**

Received 26 February 2026; revised 19 April 2026; accepted 7 July 2026. Date of publication 13 July 2026; date of current version 28 July 2026. This work was supported in part by the National Natural Science Foundation of China under Grant U25A20455, Grant 62376170, and Grant 62303287, in part by the Natural Science Foundation of Liaoning Province of China under Grant 2025-MS-013, in part by the Natural Science Foundation of Shandong Province under Grant ZR2024QF017, and in part by the Youth Innovation Technology Project of Higher School in Shandong Province under Grant 2025KJH087. Associate Editor: S.-Y. Hsieh. *(Corresponding author: Xing Wang.)*

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Digital Object Identifier 10.1109/TR.2026.3712092

I. Introduction

DISTRIBUTED optimization problems (DOPs) have become a fundamental tool for solving large-scale problems in networked systems, such as multiagent coordination, smart grids, and resource allocation [1], [2], [3], [4], [5]. Compared with centralized optimization, distributed approaches enable each agent to process local information and communicate only with its neighbors, thereby improving scalability, robustness, and privacy preservation [6], [7], [8]. Consequently, various distributed optimization frameworks have been developed, including gradient-based schemes, consensus-based algorithms, and primal-dual methods [9], [10]. In particular, gradient-based distributed methods have been extensively investigated due to their simplicity and low communication burden [11]. Moreover, recent studies have extended distributed optimization to scenarios with time-varying topologies and event-triggered communication mechanisms, significantly enhancing its applicability in real-world cyber-physical systems [12]. These advancements have laid a solid theoretical foundation for the design of distributed control and learning algorithms under realistic network constraints.

For nonlinear multiagent systems (MASs), especially those with high-order dynamics, the backstepping-based distributed optimization design typically relies on continuous information exchange among neighboring agents [13], [14], [15]. To alleviate the communication burden inherent in continuous-transmission schemes, increasing attention has been paid to event-triggered communication architectures in MASs [16], [17], [18]. Under dynamic event-triggered mechanisms (DETM), agents transmit information only when certain state-dependent or error-dependent conditions are violated, rather than at periodic or continuous instants [19], [20]. The DETM was first introduced by embedding an internal dynamic variable into the triggering rule [21]. Subsequently, Yao and Wei [22] investigated continuous-time networked control systems under stochastic impulses and proposed a DETM with adaptively adjusted thresholds. However, most existing DETM-related results mainly focus on linear systems or consensus problems [23], whereas distributed optimization for high-order uncertain nonlinear MASs under DETM remains much less explored.

Recently, universal approximators have been widely used in nonlinear control to cope with unknown dynamics and

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reduce model dependence. In particular, adaptive fault-tolerant control (FTC) has shown strong capability in compensating actuator malfunctions and external disturbances in uncertain nonlinear systems. For example, adaptive FTC schemes based on fuzzy logic systems and Nussbaum-type functions have been developed for unknown multivariable nonlinear systems with external disturbances, unknown control directions, and state-dependent actuator failures [24]. This line of research was further extended to nonlinear systems with more general fault scenarios, including multiple-fault settings [25]. In particular, for MIMO nonlinear systems with simultaneous multiplicative sensor faults and nonaffine state-dependent actuator faults, a Butterworth low-pass filter was introduced to remove the algebraic-loop difficulty arising in actuator-fault approximation [26], [27]. In addition, particle swarm optimization (PSO)-assisted fuzzy FTC studies further showed that online parameter tuning and filter-assisted fault approximation can effectively improve transient performance in practical applications such as quadrotor attitude control [28], [29].

Beyond general fault accommodation, output-feedback and observer/filter-based designs have become increasingly important for nonlinear systems with unavailable state measurements and complex operating conditions. In this direction, finite-time output-feedback fuzzy FTC frameworks have been reported for MIMO nonlinear systems with input saturation, state-dependent actuator faults, external disturbances, and unavailable state measurements [30]. Related observer- and filter-based nonlinear control designs have also been developed for systems with partial measurements and enhanced transient-performance requirements. For instance, a composite observer-based predefined-time control scheme has been proposed for robotic manipulators under deception attacks and full-state constraints, where an improved fuzzy observer, an extended state observer, and a nonsingular predefined-time filter were integrated to enhance state-estimation accuracy and transient performance [31]. These results indicate that observer/filter mechanisms provide effective tools for handling unmeasurable states, lumped uncertainties, and signal-processing difficulties in advanced nonlinear control systems.

In the multiagent domain, actuator-fault accommodation has also attracted growing attention. Recent results have shown that MAS control objectives can still be achieved in the presence of actuator faults, for example, through optimal fixed-time control architectures that explicitly incorporate actuator-fault compensation into the controller design [32]. However, these studies mainly focus on tracking, consensus, or optimal regulation problems, rather than distributed optimization under output-feedback and event-triggered communication.

Despite the aforementioned progress, the problem considered in this article remains insufficiently studied for at least three reasons. First, a large portion of the existing fuzzy FTC results concern trajectory tracking or regulation of single nonlinear plants, rather than distributed optimization over networked high-order MASs. Second, although output-feedback FTC and observer-based fuzzy designs have been reported, they do not address the coupling between distributed optimal reference generation and adaptive tracking control under event-triggered

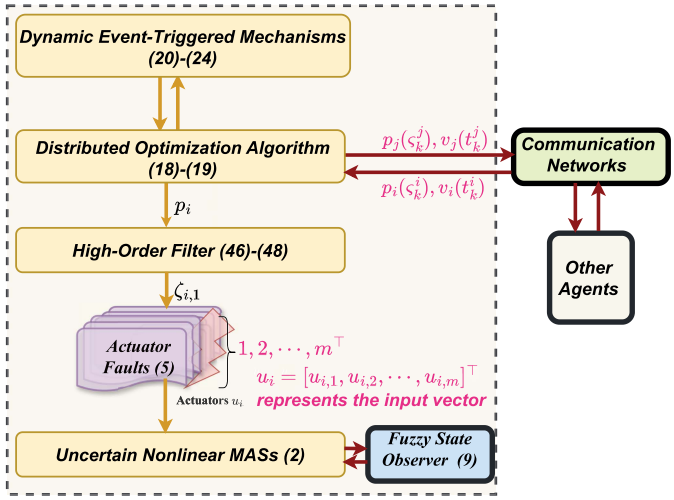


Fig. 1. Framework of the developed method.

communication [31]. Third, under DETM, the exchanged optimization signals are updated only at discrete triggering instants, which generally makes their higher order derivatives unavailable in the classical sense. This nonsmoothness creates an intrinsic obstacle to backstepping-based design for high-order nonlinear MASs, where differentiable virtual control signals are usually required [33], [34]. Therefore, the integrated treatment of distributed optimization, output-feedback state estimation, actuator-fault compensation, and dynamic event-triggered communication for high-order uncertain nonlinear MASs is still largely open.

Motivated by these observations, this article investigates distributed adaptive optimization for high-order nonlinear MASs with unknown dynamics, actuator faults, and output-only measurements under a dynamic event-triggered communication mechanism. A unified design framework is developed by combining a distributed optimization law, a high-order smoothing filter, a fuzzy state observer, and an adaptive fault-tolerant backstepping controller. The overall framework of the proposed method is illustrated in Fig. 1. In this way, the proposed method simultaneously addresses optimal reference generation, state estimation, trigger-induced signal nonsmoothness, and actuator-fault compensation within a unified Lyapunov-based framework. Distinct from prior studies, the key methodology and contributions of this article are highlighted as follows.

1) Few existing results have addressed the integrated problem of output-feedback fault-tolerant distributed optimization for high-order nonlinear MASs under dynamic event-triggered communication. To tackle this problem, a unified hierarchical design framework is developed, consisting of a distributed optimization algorithm with event-triggered communication, a high-order smoothing filter, a fuzzy state observer, and an adaptive fault-tolerant backstepping controller with actuator-fault compensation.
2) Since the higher order time derivatives of the event-triggered optimization reference $p_i(t)$ do not exist in the classical sense at the asynchronous triggering instants

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$\{t_i^k\}$ and $\{\varsigma_i^k\}$, the distributed optimization signals cannot be directly used in a standard high-order backstepping design. To overcome this difficulty, a high-order dynamic filter is introduced to generate smooth reference signals, thereby enabling controller design and stability analysis to be carried out in a continuous-time framework.

3) Compared with existing distributed optimization strategies that require full-state availability, the proposed method explicitly addresses unmeasurable states and unknown nonlinearities through an observer-based adaptive fuzzy design [35], [36]. Different from modular FTC or sequential optimization–tracking schemes, the proposed approach integrates distributed optimal reference generation, output-feedback adaptive tracking, and actuator-fault compensation into a unified Lyapunov-based analysis.

II. Preliminaries and System Descriptions

A. Notations and Graph Theory

Notations: Throughout this work, $\mathbb{R}$ stands for the real number domain. The notation $|\cdot|$ denotes the Euclidean norm for vectors and the induced 2-norm for matrices. For a symmetric matrix G, its eigenvalues are ordered as $\lambda_1(G) \le \lambda_2(G) \le \cdots \le \lambda_n(G)$. Define $U_1 = [1, 1, \ldots, 1]^{\mathrm{T}}/\sqrt{N}$, and construct an orthogonal matrix $U = [U_1, U_2]$ accordingly.

Graph Theory: We investigate a multiagent network comprising N agents, where the information exchange is modeled via an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$. In this framework, $\mathcal{V} = \{1, \ldots, N\}$ is the vertex set, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set, and $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N\times N}$ represents the weighted adjacency matrix. The weights are defined such that $a_{ij} > 0$ whenever $(i, j) \in \mathcal{E}$, and $a_{ij} = 0$ otherwise. Since the graph is undirected, the adjacency matrix is symmetric, meaning $a_{ij} = a_{ji}$. For agent i, the neighbor set is denoted by $\mathcal{N}_i = \{j \in \mathcal{V} \mid (i, j) \in \mathcal{E}\}$. The Laplacian matrix associated with $\mathcal{G}$ is defined as $L = [l_{ij}] \in \mathbb{R}^{N\times N}$, where $l_{ii} = \sum_{j\in\mathcal{N}_i} a_{ij}$ and $l_{ij} = -a_{ij}$ for $i \ne j$. It is well known that L is symmetric and positive semidefinite, and satisfies $L\mathbf{1}_N = \mathbf{0}_N$, where $\mathbf{1}_N$ is an N-dimensional vector of ones. If $\mathcal{G}$ is connected, then $\lambda_1(L) = 0$ is a simple eigenvalue, and all other eigenvalues satisfy $\lambda_i(L) > 0$ for $i = 2, \ldots, N$.

B. Fuzzy Logic Systems

To approximate the unknown continuous nonlinear dynamics $\varphi(x)$ on a compact domain $\mathcal{K}$, fuzzy logic systems are employed. A fuzzy logic system consists of M fuzzy IF–THEN rules expressed as R_i: *If* x_1 *is* C_1^i, …, x_n *is* C_n^i, *then* χ *is* G^i, where $i = 1, \ldots, M$, $x = [x_1, \ldots, x_n]^T \in \mathbb{R}^n$ and $\chi \in \mathbb{R}$ denote the input and output, respectively, and C_j^i and G^i represent fuzzy sets defined on $\mathbb{R}$. The fuzzy logic system output is given by $\chi(x) = \frac{\sum_{i=1}^M \theta_i \prod_{j=1}^n \mu_{C_j^i}(x_j)}{\sum_{i=1}^M \prod_{j=1}^n \mu_{C_j^i}(x_j)}$, where $\mu_{C_j^i}(\cdot)$ denotes the membership function and ω_i is the consequent parameter. Defining the normalized fuzzy basis functions as $\mathcal{S}_i(x) = \frac{\prod_{j=1}^n \mu_{C_j^i}(x_j)}{\sum_{l=1}^M \prod_{j=1}^n \mu_{C_j^l}(x_j)}$ and $\mathcal{S}(x) = [\mathcal{S}_1(x), \ldots, \mathcal{S}_M(x)]^T$, the fuzzy logic system can be compactly expressed as $\chi(x) = \theta^T\mathcal{S}(x)$ with $0 < \mathcal{S}^T(x)\mathcal{S}(x) \le 1$.

Lemma 1. *[37]:* For any continuous function $\varphi(x)$ defined on $\mathcal{K}$, there exists a fuzzy logic system such that

$$\sup_{x\in\mathcal{K}} \left|\varphi(x) - \theta^T\mathcal{S}(x)\right| \le H \tag{1}$$

where $H > 0$ is a constant approximation bound.

C. System Descriptions

Consider the ith follower agent with the following dynamics:

$$\begin{aligned} \dot{x}_{i,k} &= x_{i,k+1} + \varphi_{i,k}\left(\bar{x}_{i,k}\right) \\ \dot{x}_{i,n_i} &= \varpi^T u_i + \varphi_{i,n_i}\left(\bar{x}_{i,n_i}\right) \\ y_i &= x_{i,1}, \quad k = 1, \ldots, n_i - 1 \end{aligned} \tag{2}$$

where $i = 1, \ldots, N$, $\bar{x}_{i,k} = [x_{i,1}, x_{i,2}, \ldots, x_{i,k}]^T \in \mathbb{R}^k$, and $x_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,n_i}]^T \in \mathbb{R}^{n_i}$ represents the system states of the ith follower. $y_i \in \mathbb{R}$ corresponds to the system output, and $u_i = [u_{i,1}, u_{i,2}, \ldots, u_{i,m}]^T \in \mathbb{R}^m$ represents the input vector that may undergo actuator faults such as lock-in-place or loss of effectiveness during operation. The parameter vector $\varpi = [\varpi_1, \varpi_2, \ldots, \varpi_m]^T \in \mathbb{R}^m$, with each component ϖ_j $(j = 1, 2, \ldots, m)$, is assumed to be known. Furthermore, $\varphi_{i,k}(\bar{x}_{i,k}) : \mathbb{R}^k \to \mathbb{R}$ and $\varphi_{i,n_i}(x_i) : \mathbb{R}^{n_i} \to \mathbb{R}$ denote unknown nonlinear functions.

The actuator fault model is categorized into two types.

1) Actuator lock-in-place fault model

$$u_j(t) = \bar{u}_j, t \ge t_j, j \in \{j_1, j_2, \ldots, j_q\} \subset \{1, 2, \ldots, m\} \tag{3}$$

where $\bar{u}_j$ denotes the locked value of the jth actuator, while t_j signifies the specific instant of the fault onset.

2) Actuator loss-of-effectiveness fault model

$$\begin{aligned} u_j(t) = \rho_j\nu_j(t), t \ge t_j,\ j \in \overline{\{j_1, j_2, \ldots, j_q\}} \\ \cap \{1, 2, \ldots, m\} \end{aligned}$$

$$\rho_j \in [\underline{\rho_j}, 1], \quad 0 < \underline{\rho_j} \le 1 \tag{4}$$

where $\nu_j(t)$ is the actual control signal applied to the jth actuator, t_j is the fault occurrence time, and ρ_j represents the effectiveness factor of actuator $u_j(t)$. The value $1 - \rho_j$ characterizes the fault severity, and $\underline{\rho_j}$ is the lower bound of ρ_j. In particular, $\underline{\rho_j} = 1$ indicates that the corresponding actuator u_j operates normally without any fault.

Combining the lock-in-place fault model (3) and the loss-of-effectiveness fault model (4), the actual actuator input can be compactly expressed as

$$u(t) = \rho\nu(t) + \sigma(\bar{u} - \rho\nu(t)) \tag{5}$$

where $\nu(t) = [\nu_1(t), \nu_2(t), \ldots, \nu_m(t)]^T$, $\bar{u} = [\bar{u}_1, \bar{u}_2, \ldots, \bar{u}_m]^T$, $\rho = \mathrm{diag}\{\rho_1, \rho_2, \ldots, \rho_m\}$, and $\sigma = \mathrm{diag}\{\sigma_1, \sigma_2, \ldots, \sigma_m\}$. Here,

$$\sigma_j = \begin{cases} 1, & \text{if the } j\text{th actuator is locked} \\ & u_j = \bar{u}_j,\ j = 1, \ldots, m \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

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where the actual control signals are assumed to satisfy $\nu_j = r_j(y)u_0$, with $r_j(y)$ being a known positive function. The two actuator-fault models considered here admit clear engineering interpretations. Specifically, the lock-in-place fault corresponds to an actuator being stuck at a fixed output level, whereas the loss-of-effectiveness fault characterizes partial degradation, where only a fraction of the commanded control effort can be realized in practice.

Remark 1: The initial values $\varrho_{v,i}(0)$ and $\varrho_{p,i}(0)$ mainly affect the communication behavior in the initial phase. Larger initial values produce more conservative initial thresholds, which usually reduce the triggering rate at the beginning but may slow down the early suppression of distributed mismatch. By contrast, smaller initial values make the DETM more active in the transient stage and can accelerate the attenuation of initial disagreement, at the price of increased early communication. Hence, $\varrho_{v,i}(0)$ and $\varrho_{p,i}(0)$ provide a practical means of tuning the tradeoff between the initial communication burden and the transient convergence speed.

In this work, we develop a novel category of distributed optimization schemes for system (4), ensuring that all closed-loop signals stay bounded while the agent outputs converge toward the global minimizer of the optimization problem described as follows:

$$\min_{r\in\mathbb{R}} g(r), \quad g(r) = \sum_{i\in\mathcal{N}} g_i(r) \tag{7}$$

where $g(r)$ denotes the global objective function, and each $g_i : \mathbb{R} \to \mathbb{R}$, assumed to be differentiable, represents the local objective function associated with agent i. Let y_* denote a global minimizer of g. The design goal is to guarantee that

$$\lim_{t\to\infty} |y_i(t) - y_*| \le \varepsilon \tag{8}$$

where ε_i is a positive constant.

In contrast to existing distributed optimization studies that rely on prior knowledge of the gradient of local objective functions [38], our formulation considers the case where each agent utilizes the measured gradient value $\nabla g_i(y_i(t))$, which explicitly depends on its instantaneous output $y_i(t)$.

Assumption 1: g_i is ω-strongly convex and ∇g_i satisfies the ϑ-Lipschitz.

Assumption 2: For the nonlinear system described in (2), at most $m-1$ actuators are subject to stuck failures, and the aggregate effective control gain satisfies

$$\sum_{j\notin\{j_1,\ldots,j_q\}} \tau_j \varpi_j r_j(y(t)) \ge \underline{g} > 0 \qquad \forall t \ge 0.$$

Assumption 3: The undirected graph $\mathcal{G}$ is connected.

Assumption 4: For each $i \in \{1,\ldots,N\}$ and $\xi \in \{1,\ldots,n_i\}$, the unknown nonlinear function $\phi_{i,\xi}(\cdot)$ is locally Lipschitz on the compact set of interest. That is, there exists a positive constant $L_{i,\xi}$ such that

$$\left|\phi_{i,\xi}(\bar{x}_{i,\xi}) - \phi_{i,\xi}(\hat{\bar{x}}_{i,\xi})\right| \le L_{i,\xi}\|\bar{x}_{i,\xi} - \hat{\bar{x}}_{i,\xi}\| \le L_{i,\xi}\|\tilde{x}_i\|.$$

III. Main Results

To address the DOPs of uncertain nonlinear MASs, this article constructs a novel two-stage optimization algorithm. The controller is constructed via the backstepping design approach, and the stability of the system is rigorously proven.

A. Design of State Observer

According to *Lemma 1*, the fuzzy logic system $\omega_{i,\xi}^T\mathcal{S}(\hat{\bar{x}}_{i,\xi})$ is used to approximate the unknown function $\varphi_{i,\xi}(\hat{\bar{x}}_{i,\xi})$. We obtain $\varphi_{i,\xi}(\hat{\bar{x}}_{i,\xi}) = \theta_{i,\xi}^T\mathcal{S}(\hat{\bar{x}}_{i,\xi}) + H_{i,\xi}$, where $H_{i,\xi}$ is the estimation error that satisfies $|H_{i,\xi}| \le H_{iM}$ with H_{iM} being a positive constant. To estimate the unmeasurable system states, the state observer is designed as

$$\dot{\hat{x}}_i = A_i\hat{x}_i + \sum_{\xi=1}^{n_i} b_{i,\xi}\hat{\varphi}_{i,\xi}\left(\hat{\bar{x}}_{i,\xi} \mid \hat{w}_{i,\xi}\right) + K_i y_i + B_i\varpi u_i \tag{9}$$

where $\hat{\bar{x}}_{i,\xi} = (\hat{x}_{i,1},\ldots,\hat{x}_{i,\xi})^T$ is the estimate of $\bar{x}_{i,\xi}$, $\hat{\omega}_{i,\xi}$ is the estimate of $\omega_{i,\xi}$, $\hat{\varphi}_{i,\xi}(\hat{\bar{x}}_{i,\xi}|\hat{\omega}_{i,\xi}) = \hat{\omega}_{i,\xi}^T\mathcal{S}(\hat{\bar{x}}_{i,\xi})$, $\bar{x}_{i,\xi} = [x_{i,1},\ldots,x_{i,\xi}]^T$, $b_{i,\xi} = [0,\ldots,0,1,\ldots,0]^T$, $B_i = [0,\ldots,0,1]^T$, $K_i = [l_{i,1}, l_{i,2},\ldots,l_{i,n_i}]^T$, and the matrix A_i is expressed as

$$A_i = \begin{bmatrix} -l_{i,1} & 1 & 0 & \cdots & 0 \\ -l_{i,2} & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & I & 0 \\ -l_{i,n_i} & 0 & 0 & \cdots & 0 \end{bmatrix} \tag{10}$$

where $l_{i,\xi}(\xi = 1,\ldots,n_i)$ are design parameters chosen such that the matrix A_i is Hurwitz. For any given matrix $Q_i > 0$, one can always find a symmetric positive definite matrix $P_i = P_i^T > 0$ satisfying $P_iA_i + A_i^TP_i = -2Q_i$.

Define the estimation error vector $\tilde{x}_i$ as $\tilde{x}_i = x_i - \hat{x}_i$, where $x_i = [x_{i,1},\ldots,x_{i,n_i}]^T$ and $\hat{x}_i = [\hat{x}_{i,1},\ldots,\hat{x}_{i,n_i}]^T$. From (2) and (9), we obtain

$$\dot{\tilde{x}}_i = A_i\tilde{x}_i + \sum_{\xi=1}^{n_i}\omega_{i,\xi}^T\Delta\varphi_{i,\xi} + \sum_{\xi=1}^{n_i} b_{i,\xi}\tilde{\theta}_{i,\xi}^T\mathcal{S}(\hat{x}_{i,\xi}) + H_i \tag{11}$$

where $\tilde{\theta}_{i,\xi} = \theta_{i,\xi} - \hat{\theta}_{i,\xi}$, $H_i = [H_{i,1},\ldots,H_{i,n_i}]^T$, and $\Delta\varphi_{i,\xi} = [\varphi_{i,\xi}(\bar{x}_{i,\xi}) - \varphi_{i,\xi}(\hat{\bar{x}}_{i,\xi})]^T$.

The Lyapunov function is chosen as

$$V_{xi} = \tilde{x}_i^TP_i\tilde{x}_i. \tag{12}$$

Differentiating V_{xi}, one has

$$\dot{V}_{xi} = -\tilde{x}_i^TQ\tilde{x}_i + \tilde{x}_i^TP_i\sum_{\xi=1}^{n_i}\theta_{i,\xi}^T\Delta\varphi_{i,\xi} + \sum_{\xi=1}^{n_i}\tilde{x}_i^TP_i\tilde{\theta}_{i,\xi}^TS\left(\bar{x}_{i,\xi}\right) + \tilde{x}_i^TP_iH_i. \tag{13}$$

From Young's inequality and Assumption 4, the following inequality holds:

$$\tilde{x}_i^TP_i\sum_{\xi=1}^{n_i} b_{i,\xi}\tilde{\theta}_{i,\xi}^T\mathcal{S}(\hat{\bar{x}}_{i,\xi}) \le \frac{1}{2}|\tilde{x}_i|^2 + \frac{1}{2}|P_i|^2\sum_{\xi=1}^{n_i}\tilde{\theta}_{i,\xi}^T\tilde{\theta}_{i,\xi} \tag{14}$$

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$$\tilde{x}_i^\top P_i \sum_{\xi=1}^{n_i} \theta_{i,\xi}^\top \Delta\phi_{i,\xi} \le c_{i\phi}\|\tilde{x}_i\|^2 \tag{15}$$

$$\tilde{x}_i^T P_i H_i \le \frac{1}{2}|\tilde{x}_i|^2 + \frac{1}{2}|P_i|^2|H_{iM}|^2 \tag{16}$$

where $c_{i\phi} > 0$ is a constant determined by Assumption 4 on the compact set of interest.

Then, substituting (14)–(16) into (13) yields

$$\dot{V}_{xi} \leqslant -a_{xi}|\tilde{x}_i|^2 + |P_i|^2 \sum_{\xi=1}^{n_i} \frac{1}{2}\tilde{\theta}_{i,\xi}^T\tilde{\theta}_{i,\xi} + \tau_{xi} \tag{17}$$

where $a_{x_i} = \lambda_{\min}(Q_i) - c_{i\phi} - 2 > 0$ and $\tau_{xi} = |P_i|^2 |H_{iM}|^2 + \frac{1}{2}|P_i|^2 \sum_{\xi=1}^{n_i} \theta_{i,\xi}^T\theta_{i,\xi}$.

B. Distributed Optimization Algorithm Under DETC

In this study, interagent communication is governed by a DETM, where information exchange is activated only when certain triggering conditions are satisfied rather than at uniform time intervals. Consequently, the communication process is asynchronous and aperiodic, as transmissions depend on the current states of individual agents. Based on such a communication strategy, the distributed optimization scheme utilizing the dynamic event-triggered control (DETC) framework is formulated as follows:

$$\begin{aligned}\dot{p}_i = &-\alpha_1 \nabla g_i(y_i) - \alpha_2 \sum_{j=1}^{N} a_{ij}(p_i(\varsigma_k^i) - p_j(\varsigma_{k'}^j)) \\ &- \alpha_3 \sum_{j=1}^{N} a_{ij}(v_i(t_k^i) - v_j(t_{k'}^j)), \quad i = 1, 2, \ldots, N\end{aligned} \tag{18}$$

$$\dot{v}_i = \alpha_4 \sum_{j=1}^{N} a_{ij}(p_i(\varsigma_k^i) - p_j(\varsigma_{k'}^j)) \tag{19}$$

where p_i and v_i denote the auxiliary variables introduced in the distributed optimization framework, with p_i acting as the reference signal of agent i. The positive design parameters $\alpha_1, \alpha_2, \alpha_3$, and α_4 specify the update gains of the algorithm. The transmission instants $\{t_k^i\}_{k\in\mathbb{N}}$ and $\{\varsigma_k^i\}_{k\in\mathbb{N}}$, initialized by $t_0^i = \varsigma_0^i = 0$, correspond to the event-triggered sequences (ETSs), at which $v_i(t_k^i)$ and $p_i(\varsigma_k^i)$ are communicated to neighboring agents. For any neighbor j, $t_{k'}^j = \max\{t_k^j \mid t_k^j \le t\}$ and $\varsigma_{k'}^j = \max\{\varsigma_k^j \mid \varsigma_k^j \le t\}$ denote its most recent triggering instants.

Remark 2: Two independent event-triggered sequences are introduced for the variables v_i and p_i because these two signals play different roles in the distributed optimization layer and may evolve on different time scales. Using separate triggering rules allows each communication channel to be updated only when its own mismatch becomes significant, which can avoid unnecessary transmissions that may arise under a unified trigger. The price paid for this flexibility is a higher local monitoring complexity, since two triggering conditions must be checked in parallel.

Based on this distributed optimization law, we next specify the DETMs that generate the two communication sequences.

C. DETMs Design

The ETSs $\{t_k^i\}_{k\in\mathbb{N}}$ and $\{\varsigma_k^i\}_{k\in\mathbb{N}}$ are, respectively, generated by the following DETMS:

$$t_{k+1}^i = \inf\{t > t_k^i \mid \phi_{v,i}(t) \ge 0\} \tag{20}$$

$$\varsigma_{k+1}^i = \inf\{t > \varsigma_k^i \mid \phi_{p,i}(t) \ge 0\} \tag{21}$$

where the triggering functions $\phi_{v,i}(\cdot)$ and $\phi_{p,i}(\cdot)$ are defined, respectively, as

$$\begin{cases}\phi_{v,i} = a_v|e_{v,i}|^2 - h_{v,i}\varrho_{v,i} - b_v\left|\sum_{j=1}^{N} a_{ij}\left(\tilde{v}_j(t) - \tilde{v}_i(t)\right)\right|^2 \\ [6pt]\phi_{p,i} = a_p|e_{p,i}|^2 - h_{p,i}\varrho_{p,i} - b_p\left|\sum_{j=1}^{N} a_{ij}\left(\tilde{p}_j(t) - \tilde{p}_i(t)\right)\right|^2\end{cases} \tag{22}$$

where $e_{v,i} = \check{v}_i(t) - v_i(t)$ and $\check{v}_i(t) = v_i(t_k^i)$ for $t \in [t_k^i, t_{k+1}^i)$, while $e_{p,i} = \check{p}_i(t) - p_i(t)$ and $\check{p}_i(t) = p_i(\varsigma_k^i)$ for $t \in [\varsigma_k^i, \varsigma_{k+1}^i)$. Here, $a_v, b_v, h_{v,i}, a_p, b_p, h_{p,i} > 0$ are design parameters, and a_{ij} denotes the (i,j)th entry of the adjacency matrix of the communication graph.

The auxiliary variables $\varrho_{v,i}$ and $\varrho_{p,i}$ are introduced to dynamically adjust the triggering thresholds. Their evolutions are defined as

$$\dot{\varrho}_{v,i} = b_v|\sum_{j=1}^{N} a_{ij}(\tilde{v}_j(t) - \tilde{v}_i(t)|^2 - f_{v,i}\varrho_{v,i} - a_v|e_{v,i}|^2 \tag{23}$$

$$\dot{\varrho}_{p,i} = b_p|\sum_{j=1}^{N} a_{ij}(\tilde{p}_j(t) - \tilde{p}_i(t))|^2 - f_{p,i}\varrho_{p,i} - a_p|e_{p,i}|^2 \tag{24}$$

where $a_v, b_v, h_{v,i}, f_{v,i}, a_p, b_p, h_{p,i}, f_{p,i} > 0$ and the initial conditions $\varrho_{v,i}(0), \varrho_{p,i}(0) > 0$ are positive design constants to be specified subsequently. For the ith agent, the transmitted information during communication is given by $\tilde{v}_i(t) = v_i(t_k^i)$ and $\tilde{p}_i(t) = p_i(\varsigma_k^i)$, respectively. For the sake of notational brevity, the argument t is dropped in the subsequent derivations.

Remark 3: The parameters in the proposed DETM determine a tradeoff among communication frequency, transient mismatch attenuation, and the ultimate optimization/tracking accuracy. In particular, larger values of a_v and a_p increase the sensitivity of the triggering conditions to the local sampling errors $e_{v,i}$ and $e_{p,i}$, which tends to activate communication more frequently and suppress error accumulation more aggressively. The coefficients b_v and b_p regulate the influence of the neighboring disagreement terms, and therefore, affect how strongly the triggering mechanism reacts to distributed coordination mismatch. Moreover, $h_{v,i}$ and $h_{p,i}$ affect the threshold terms in the triggering functions, whereas $f_{v,i}$ and $f_{p,i}$ govern the decay rates of the auxiliary dynamic variables $\varrho_{v,i}$ and $\varrho_{p,i}$. Conservative choices generally reduce communication activity but may lead to slower transient attenuation and a larger ultimate residual set, whereas aggressive choices improve performance at the expense of more frequent transmissions. Hence, in practice, these parameters should be

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selected by balancing the desired communication saving against the acceptable residual optimization error.

The tracking error states are defined as

$$\tilde{y} = y - p, \quad \tilde{p} = p - y_0, \quad \tilde{v} = v - v_0 \tag{25}$$

in which $s_1 = [s_{1,1}, \ldots, s_{N,1}]^T$, $y_i = [y_1, \ldots, y_N]^T$, $y_0 = [y^*, \ldots, y^*]^T \in \mathbb{R}^N$, and v_0 is defined as

$$v_0 = U_1 U_1^T v(0) - \frac{c_1}{c_3} U_2 (U_2^T \mathcal{L} U_2)^{-1} U_2^T \Delta_g(y_0). \tag{26}$$

Due to the orthogonality of U_1, $U_1^T \mathcal{L} = 0$, $\mathcal{L} U_1 = 0$, and $U_1^T \Delta_g(y_0) = 0$ can be obtained from Assumptions 1.

Then, the following formula can be inferred:

$$\begin{aligned} c_3 \mathcal{L} v_0 &= c_3 (U_1 U_1^T + U_2 U_2^T) \mathcal{L} v_0 = -c_1 U_2 U_2^T \Delta_g(y_0) \\ &= -c_1 (I_n - U_1 U_1^T) \Delta_g(y_0) = -c_1 \Delta_g(y_0) \end{aligned} \tag{27}$$

where $U_\otimes = [U_1, U_2]$. Then, (18) and (19) can be expressed as

$$\dot{\tilde{p}} = -\alpha_1 \tilde{\nabla}_g(y, p^*) - \alpha_2 \mathcal{L} \tilde{p} - \alpha_3 \mathcal{L} \tilde{v} - \alpha_2 \mathcal{L} e_p - \alpha_3 \mathcal{L} e_v \tag{28}$$

$$\dot{\tilde{v}} = \alpha_4 \mathcal{L} \tilde{p} + \alpha_4 \mathcal{L} e_p \tag{29}$$

where $\tilde{\nabla}_g(y, p^*) = \nabla_g(y) - \nabla_g(p^*)$, with the compact vector notations given by $\nabla_g(y) = \mathrm{col}\{\nabla g_1(y_1), \ldots, \nabla g_N(y_N)\}$, $e_p = \mathrm{col}\{e_{p,1}, \ldots, e_{p,N}\}$, and $e_v = \mathrm{col}\{e_{v,1}, \ldots, e_{v,N}\}$.

D. Lyapunov Function for the $(\tilde{p}_i, \tilde{v}_i)$-Subsystem

Combining the DETMs (20)–(21) with the dynamic equations of $\varrho_{q,i}$ given in (23) and $\psi_{p,i}$ in (24), one obtains

$$\begin{aligned} \dot{\varrho}_{p,i}(t) &\geq -(f_{p,i} + h_{p,i}) \varrho_{p,i}(t) \\ \dot{\varrho}_{v,i}(t) &\geq -(f_{v,i} + h_{v,i}) \varrho_{v,i}(t) \end{aligned} \tag{30}$$

which implies

$$\begin{aligned} \varrho_{p,i}(t) &\geq e^{-(f_{p,i}+h_{p,i})t} \varrho_{p,i}(0) > 0, \quad t \geq 0 \\ \varrho_{v,i}(t) &\geq e^{-(f_{v,i}+h_{v,i})t} \varrho_{v,i}(0) > 0, \quad t \geq 0 \end{aligned} \tag{31}$$

provided that the initial values satisfy $\varrho_{p,i}(0) > 0$ and $\varrho_{v,i}(0) > 0$. Accordingly, the Lyapunov function candidate is selected as

$$V_\xi(\xi) = \frac{1}{2} \xi^T \left(\begin{bmatrix} k_1 & k_2 \\ k_2 & k_3 \end{bmatrix} \otimes I_N \right) \xi + \sum_{i=1}^{N} (\varrho_{p,i} + \varrho_{v,i}) \tag{32}$$

where $\xi = [\hat{p}^T, \hat{v}^T]^T$. $k_1, k_2, k_3 > 0$ satisfy

$$\begin{aligned} &k_1 k_3 - k_2^2 > 0 \\ &k_2 \alpha_2 - k_3 \alpha_4 = k_1 \alpha_3. \end{aligned} \tag{33}$$

Taking the derivative of V_ξ along (28), (29), (30), ad (32) yields

$$\begin{aligned} \dot{V}_\xi = &- \xi^T Q_2 \xi - k_1 \alpha_1 \hat{p}^T \tilde{\nabla}_g(p, p^*) - k_1 \alpha_1 \hat{p}^T \tilde{\nabla}_g(y, p) \\ &- k_2 \alpha_1 \hat{v}^T \tilde{\nabla}_g(y, p^*) - \alpha_5 k_2 \alpha_3 \hat{v}^T \mathcal{L} \hat{v} - 2 \xi^T Q_3 e_\xi \\ &+ \sum_{i=1}^{N} \left(-f_{p,i} \psi_{p,i} - f_{v,i} \varrho_{v,i} + b_p \left| \sum_{j=1}^{N} a_{ij} (\tilde{p}_j - \tilde{p}_i) \right|^2 \right. \\ &\left. + b_v \left| \sum_{j=1}^{N} a_{ij} (\tilde{v}_j - \tilde{v}_i) \right|^2 - a_p |e_{p,i}|^2 - a_v |e_{v,i}|^2 \right) \end{aligned} \tag{34}$$

where $\alpha_5 \in (0, 1)$, and matrices are

$$Q_2 = \begin{bmatrix} k_1 \alpha_2 - k_2 \alpha_4 & \frac{1}{2} k_1 \alpha_3 + \frac{1}{2} k_2 \alpha_2 - \frac{1}{2} k_3 \alpha_4 \\ * & (1 - \alpha_5) k_2 \alpha_3 \end{bmatrix} \otimes \mathcal{L} \tag{35}$$

$$Q_3 = \frac{1}{2} \begin{bmatrix} k_1 \alpha_2 - k_2 \alpha_4 & k_1 \alpha_3 \\ k_2 \alpha_2 - k_3 \alpha_4 & k_2 \alpha_3 \end{bmatrix} \otimes \mathcal{L}. \tag{36}$$

By applying Young's inequality alongside the definition of $\tilde{\nabla}_c$, we derive

$$\begin{aligned} &|\tilde{\nabla}_g(y, p^*)|^2 = |\tilde{\nabla}_g(y, p) + \tilde{\nabla}_g(p, p^*)|^2 \\ &\leq (1 + \sigma_3) |\tilde{\nabla}_g(p, p^*)|^2 + \left(1 + \frac{1}{\sigma_3}\right) |\tilde{\nabla}_g(y, p)|^2 \end{aligned} \tag{37}$$

with σ_3 being a positive constant. Furthermore, Assumption 1 implies that

$$|\tilde{\nabla}_g(p, p^*)|^2 = |\nabla_g(p) - \nabla_g(p^*)|^2 \leq \vartheta \tilde{p}^T \tilde{\nabla}_g(p, p^*) \tag{38}$$

$$|\tilde{\nabla}_g(y, p)|^2 = |\nabla_g(y) - \nabla_g(p)|^2 \leq \vartheta^2 |\tilde{y}|^2. \tag{39}$$

Combining the aforementioned inequalities results in

$$|\tilde{\nabla}_g(y, p^*)|^2 \leq (1 + \sigma_3) \vartheta \tilde{p}^T \tilde{\nabla}_g(p, p^*) + \left(1 + \frac{1}{\sigma_3}\right) \vartheta^2 |\tilde{y}|^2. \tag{40}$$

Using the aforementioned inequalities, one further obtains

$$\begin{aligned} |\tilde{p}|^2 \leq &\left(1 + \tfrac{1}{\sigma_2}\right) (\xi + e_\xi)^T Q_1 (\xi + e_\xi) \\ &+ \alpha_1^2 \vartheta^2 (1 + \sigma_2) \left(1 + \tfrac{1}{\sigma_3}\right) |\tilde{y}|^2 \\ &+ (1 + \sigma_2)(1 + \sigma_3) \alpha_1^2 \vartheta \tilde{p}^T \tilde{\nabla}_g(p, p^*). \end{aligned} \tag{41}$$

By combining (31) with (32) and (33) and applying the inequality $\omega \|\tilde{p}\|^2 \leq \tilde{p}^T \tilde{\nabla}_g(p, p^*)$ from Assumption 1, we obtain

$$\dot{V}_\xi \leq - \begin{bmatrix} \xi^T & e_\xi^T \end{bmatrix} \begin{bmatrix} \check{Q}_2 & -\check{Q}_3 \\ -\check{Q}_3 & \check{Q}_4 \end{bmatrix} \begin{bmatrix} \xi \\ e_\xi \end{bmatrix} \tag{42}$$

$$[3mm] - \sum_{i=1}^{N} (f_{p,i} \varrho_{p,i} + f_{v,i} \varrho_{v,i}) - \alpha_6 \tilde{p}^T \tilde{\nabla}_g(p, p^*) \tag{43}$$

$$- \alpha_7 |\tilde{v}|^2 + \alpha_8 |\tilde{y}|^2 \tag{44}$$

with the matrices and coefficients defined as follows:

$$\begin{aligned} \check{Q}_s &= Q_s - \dot{b} \lambda_N (l_2 \otimes \mathcal{L}), \quad s = 2, 3, 4 \\ \alpha_6 &= \left(k_1 \left(1 - \tfrac{\sigma_4}{2\omega}\right) - \tfrac{k_2 \vartheta}{2\sigma_5} (1 + \sigma_3) \right) \alpha_1 \\ \alpha_7 &= k_2 \left(\alpha_5 \alpha_3 \lambda_2 - \tfrac{1}{2} \alpha_1 \sigma_5 \right) \\ \alpha_8 &= \tfrac{\alpha_1 \vartheta^2}{2} \left(\tfrac{k_1}{\sigma_4} + \tfrac{k_2}{\sigma_5} \left(1 + \tfrac{1}{\sigma_3}\right) \right) \end{aligned} \tag{45}$$

where $\dot{b} = \max\{b_p, b_v\}$.

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E. Lyapunov Function of the $(s_{i,q}, \theta)$-Subsystem and Adaptive Fuzzy Controller Design

Subsequently, an enhanced resilient distributed observer is constructed as follows:

$$\dot{\zeta}_{i,1} = -\alpha_1 \nabla g_i(y_i) - \iota(\zeta_{i,1} - \zeta_{i,2}) \tag{46}$$

$$\dot{\zeta}_{i,2} = -\alpha_1 \nabla g_i(y_i) - \iota(\zeta_{i,2} - \zeta_{i,3}) \tag{47}$$

$$\cdots$$

$$\dot{\zeta}_{i,n_i} = -\alpha_1 \nabla g_i(y_i) - \iota(\zeta_{i,n_i} - p_i) \tag{48}$$

with p_i defined in (18) and ι being a positive constant. Analyzing the configuration of (46)–(48), it is evident that the state $\zeta_{i,1}$ evolves according to an n_ith-order filtering dynamic.

To establish the existence of high-order derivatives of the observer state $\zeta_{i,1}$, we analyze the hierarchical observer structure described in (46)–(48). From (46), the first-order derivative $\dot{\zeta}_{i,1}$ exists because its right-hand side is composed of $\zeta_{i,1}$, $\zeta_{i,2}$, and the gradient term $\nabla g_i(y_i)$. For any $n_i \geq 2$, obtaining the n_ith derivative of $\zeta_{i,1}$ requires repeatedly differentiating (46). Since $\zeta_{i,2}$ explicitly appears in $\dot{\zeta}_{i,1}$, the existence of $\zeta_{i,1}^{(n_i)}$ depends on the existence of $\zeta_{i,2}^{(n_i-1)}$. According to (47), $\dot{\zeta}_{i,2}$ exists, while the existence of its (n_i-1)th derivative depends on the (n_i-2)th derivative of $\zeta_{i,3}$. At the last layer, (48) ensures $\dot{\zeta}_{i,n_i}$ exists because its right-hand side includes $\nabla g_i(y_i)$ and the reference signal p_i, which is assumed to be differentiable. Hence, $\zeta_{i,n_i}^{(1)}$ exists, which implies that$\zeta_{i,n_i-1}^{(2)}$ exists, and by induction, all derivatives of $\zeta_{i,1}$ up to order n_i exist.

Remark 4: Note that the high-order filter guarantees the differentiability required by the backstepping design, while inevitably introducing a transient lag between p_i and the filtered signal used by the controller. Such a lag does not destroy the boundedness of the closed-loop system, but contributes to the practical residual error. Hence, the obtained result should be interpreted as practical convergence to a neighborhood of the global minimizer rather than exact asymptotic recovery of the ideal unfiltered reference.

Lemma 2. *[39]:* Given an arbitrary constant $\iota > 0$, the observer errors $\zeta_{i,j} - p_i\,(j = 1, \ldots, n_i; i = 1, \ldots, N)$ are guaranteed to vanish exponentially, while the existence of the derivatives of the observer state $\zeta_{i,1}$ up to the n_ith order is ensured.

Since $\tilde{\zeta}_{i,n_i}$ decays to zero exponentially, and considering the upper triangular configuration of the error dynamics (50)–(52), it can be inferred that the signals $\tilde{\zeta}_{i,1}, \ldots, \tilde{\zeta}_{i,n_i-1}$ also demonstrate exponential convergence to the origin. Consequently, the term $\zeta_{i,j} - p_i$ achieves exponential convergence to zero. Based on the demonstrated stability of the observer, we now proceed to design the adaptive controller to ensure satisfactory tracking performance.

Step 1: The control design proceeds as follows:

$$s_{i,1} = x_{i,1} - \kappa_i \tag{49}$$

$$\begin{aligned}\dot{s}_{i,1} &= \dot{x}_{i,1} - \dot{\kappa}_i \\ &= \hat{x}_{i,2} + \tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}^T D(\hat{x}_{i,1}) + \hat{\theta}^T D(\hat{x}_{i,1}) \\ &\quad + \epsilon_{i,1} - \dot{\kappa}_i \\ &= s_{i,2} + \alpha_{i,1} + \tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}^T D(\hat{x}_{i,1}) + \hat{\theta}^T D(\hat{x}_{i,1}) \\ &\quad + \epsilon_{i,1}.\end{aligned} \tag{50}$$

The Lyapunov function $V_{i,1}$ is chosen as

$$V_{i,1} = \frac{1}{2} s_{i,1}^2 + \frac{1}{2\gamma_{i,1}} \tilde{\theta}_{i,1}^T \tilde{\theta}_{i,1}. \tag{51}$$

Differentiating $V_{i,1}$, one has

$$\begin{aligned}\dot{V}_{i,1} &= s_{i,1}\dot{s}_{i,1} - \frac{1}{\gamma_{i,1}} \tilde{\theta}_{i,1} \dot{\hat{\theta}}_{i,1} \\ &= s_{i,1}\Big(s_{i,2} + \alpha_{i,1} + \tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}^T D(\hat{x}_{i,1}) \\ &\quad + \hat{\theta}^T D(\hat{x}_{i,1}) + \epsilon_{i,1}\Big) - \frac{1}{\gamma_{i,1}} \tilde{\theta}_{i,1} \dot{\hat{\theta}}_{i,1}.\end{aligned} \tag{52}$$

From Young's inequality, the following inequalities hold:

$$s_{i,1}\tilde{x}_{i,2} \leq \frac{1}{2} s_{i,1}^2 + \frac{1}{2} |\tilde{x}_{i,2}|^2 \leq \frac{1}{2} s_{i,1}^2 + \frac{1}{2} |\tilde{x}_i|^2 \tag{53}$$

$$[1ex] s_{i,1}\Delta\varphi_{i,1} \leq s_{i,1}\theta_{i,1}^T (D(x_{i,1}) - D(\hat{x}_{i,1})) \leq s_{i,1}^2 + |\theta_{i,1}|^2 \tag{54}$$

$$[1ex] s_{i,1}\epsilon_{i,1} \leq \frac{1}{2} s_{i,1}^2 + \frac{1}{2} |\epsilon^*|^2. \tag{55}$$

The virtual controller $\alpha_{i,1}$ is designed as

$$\alpha_{i,1} = -c_{i,1} s_{i,1} - \hat{\theta}_{i,1}^T S(\hat{x}_{i,1}) - 2 s_{i,1} \tag{56}$$

and the adaptive law is constructed as

$$\dot{\hat{\theta}}_{i,1} = \Xi_{i,1} D_{i,1}(\hat{x}_{i,1}) s_{i,1} - \rho_{i,1} \hat{\theta}_{i,1}. \tag{57}$$

Substituting (53)–(57) into $\dot{V}_{i,1}$ results in

$$\begin{aligned}\dot{V}_{i,1} &\leq -c_{i,1} s_{i,1}^2 + s_{i,1} s_{i,2} + \frac{\rho_{i,1}}{r_{i,1}} \tilde{\theta}_{i,1}^\top \tilde{\theta}_{i,1} \\ &\quad + \frac{1}{2} |\tilde{x}_{i,1}|^2 + \frac{1}{2} |\epsilon^*|^2 + |\theta_{i,1}|^2.\end{aligned} \tag{58}$$

Step z_i $(2 \leq z_i \leq n_i - 1)$**:** Following a procedure similar to Step 1, we obtain

$$s_{i,z_i} = \hat{x}_{i,z_i} - \alpha_{i,z_i-1} - \kappa_i^{(z_i-1)} \tag{59}$$

$$\begin{aligned}\dot{s}_{i,z_i} &= \dot{\hat{x}}_{i,z_i} - \dot{\alpha}_{i,z_i-1} - \kappa_i^{(z_i)} \\ &= s_{i,z_i+1} + \alpha_{i,z_i} + \hat{\theta}_{i,z_i}^T D(\hat{\bar{x}}_{i,z_i}) + l_{i,z_i}\tilde{x}_{i,z_i} - H_{i,z_i} \\ &\quad + \tilde{\theta}_{i,z_i}^T D(\hat{\bar{x}}_{i,z_i}) - \tilde{\theta}_{i,z_i}^T D(\hat{\bar{x}}_{i,z_i}) - \sum_{d=1}^{z_i-1} \frac{\partial \alpha_{i,d-1}}{\partial \kappa_i^{(d-1)}} \dot{\kappa}_{i,d} \\ &\quad - \frac{\partial \alpha_{i,z_i-1}}{\partial x_{i,1}} \left(\tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}_{i,1}^T D(\hat{x}_{i,1}) + \epsilon_{i,1}\right). \\ &\quad - \frac{1}{\gamma_{i,z_i}} \tilde{\theta}_{i,z_i} \dot{\hat{\theta}}_{i,z_i}.\end{aligned} \tag{60}$$

The auxiliary term H_{i,z_i} is defined as

$$H_{i,z_i} = -\frac{\partial \alpha_{i,z_i-1}}{\partial x_{i,1}} \left(\hat{x}_{i,2} + \hat{\theta}_{i,1}^T D(\hat{x}_{i,1})\right) - \sum_{d=1}^{z_i-1} \frac{\partial \alpha_{i,d-1}}{\partial \hat{\theta}_{i,d}} \dot{\hat{\theta}}_{i,d}$$

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$$-\sum_{d=1}^{z_i-1}\frac{\partial\alpha_{i,z_i-1}}{\partial\hat{x}_{i,d}}(\hat{x}_{i,d+1}+\hat{\theta}_{i,d}^{T}D(\hat{\bar{x}}_{i,d})+l_{i,d}\tilde{x}_{i,1}). \tag{61}$$

Consider the following Lyapunov function:

$$V_{i,z_i}=V_{i,z_i-1}+\frac{1}{2}s_{i,z_i}^{2}+\frac{1}{2\gamma_{i,z_i}}\tilde{\theta}_{i,z_i}^{T}\tilde{\theta}_{i,z_i}. \tag{62}$$

We can get the differentiation of V_{i,z_i} as

$$\begin{aligned}\dot{V}_{i,z_i}&=\dot{V}_{i,z_i-1}+s_{i,z_i}\dot{s}_{i,z_i}-\frac{1}{\gamma_{i,z_i}}\tilde{\theta}_{i,z_i}^{T}\dot{\hat{\theta}}_{i,z_i}\\&=\dot{V}_{i,z_i-1}+s_{i,z_i}\Big(s_{i,z_i+1}+\alpha_{i,z_i}+\hat{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})\\&\quad+l_{i,z_i}\tilde{x}_{i,z_i}+\tilde{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})-\tilde{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})\\&\quad-\sum_{d=1}^{z_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\kappa_i^{(d-1)}}\dot{\kappa}_{i,d}-\frac{\partial\alpha_{i,z_i-1}}{\partial x_{i,1}}(\tilde{x}_{i,2}+\Delta\varphi_{i,1}\\&\quad+\tilde{\theta}_{i,1}^{T}D(\hat{x}_{i,1})+\epsilon_{i,1}\Big)-H_{i,z_i}\Big)-\frac{1}{\gamma_{i,z_i}}\tilde{\theta}_{i,z_i}\dot{\hat{\theta}}_{i,z_i}.\end{aligned} \tag{63}$$

Using Young's inequality, we arrive at following the inequalities:

$$\begin{aligned}&-s_{i,z_i}\frac{\partial\alpha_{i,z_i-1}}{\partial x_{i,1}}\big(\tilde{x}_{i,2}+\Delta\varphi_{i,1}+\tilde{\theta}_{i,1}^{T}D(\hat{x}_{i,1})+\epsilon_{i,1}\big)\\&\le 2s_{i,z_i}^{2}\left(\frac{\partial\alpha_{i,z_i-1}}{\partial x_{i,1}}\right)^{2}+\frac{1}{2}\epsilon_{i,1}^{*2}+\frac{1}{2}\big|\tilde{x}_i\big|^{2}+\big|\theta_{i,1}\big|^{2}\\&\quad+\frac{1}{2}\tilde{\theta}_{i,1}^{T}\tilde{\theta}_{i,1}\end{aligned} \tag{64}$$

$$[1ex]-s_{i,z_i}\tilde{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})\le\pi s_{i,z_i}^{2}+\frac{1}{\pi}\tilde{\theta}_{i,z_i}^{T}\tilde{\theta}_{i,z_i}. \tag{65}$$

We define $\bar{H}_{i,z_i}=H_{i,z_i}+2s_{i,z_i}\left(\frac{\partial\alpha_{i,z_i-1}}{\partial x_{i,1}}\right)^{2}$. Therefore, we can obtain the following expression:

$$\begin{aligned}\dot{V}_{i,z_i}\le&\dot{V}_{i,z_i-1}+s_{i,z_i}\Bigg[s_{i,z_i+1}+\alpha_{i,z_i}+\tilde{\theta}_{i,z_i}^{T}D\left(\hat{\bar{x}}_{i,z_i}\right)+\bar{H}_{i,z_i}\\&+\hat{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})-l_{i,z_i}\tilde{x}_{i,z_i}-\sum_{d=1}^{z_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\kappa_i^{(d-1)}}\dot{\kappa}_{i,d}\Bigg]\\&+\frac{1}{2}\tilde{\theta}_{i,1}^{T}\tilde{\theta}_{i,1}+\pi s_{i,z_i}^{2}+\frac{1}{\pi}\tilde{\theta}_{i,z_i}^{T}\tilde{\theta}_{i,z_i}+\frac{1}{2}\epsilon_{i,1}^{*2}\\&+\frac{1}{2}\big|\tilde{x}_i\big|^{2}+\big|\theta_{i,1}\big|^{2}-\frac{1}{\gamma_{i,z_i}}\tilde{\theta}_{i,z_i}^{T}\dot{\hat{\theta}}_{i,z_i}.\end{aligned} \tag{66}$$

Then, α_{i,z_i} is designed as

$$\begin{aligned}\alpha_{i,z_i}=&-c_{i,z_i}s_{i,z_i}-s_{i,z_i-1}-\pi s_{i,z_i}-l_{i,z_i}\tilde{x}_{i,z_i}-\bar{H}_{i,z_i}\\&-\hat{\theta}_{i,z_i}^{T}D(\hat{\bar{x}}_{i,z_i})+\sum_{d=1}^{z_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\kappa_i^{(d-1)}}\dot{\kappa}_{i,d}\end{aligned} \tag{67}$$

and $\dot{\hat{\theta}}_{i,z_i}$ is constructed as

$$\dot{\hat{\theta}}_{i,z_i}=\Xi_{i,z_i}D_{i,z_i}(\hat{\bar{x}}_{i,z_i})s_{i,z_i}-\rho_{i,z_i}\hat{\theta}_{i,z_i}. \tag{68}$$

By substituting (68)–(70) into (67), we obtain

$$\begin{aligned}\dot{V}_{i,z_i}\le&-\sum_{d=1}^{z_i}\left(c_{i,d}s_{i,d}^{2}+\frac{\rho_{i,d}}{\gamma_{i,d}}\tilde{\theta}_{i,d}^{T}\tilde{\theta}_{i,d}\right)+s_{i,z_i}s_{i,z_i+1}+\frac{|\tilde{x}_{i,1}|^{2}}{2}\\&+\frac{1}{\pi}\tilde{\theta}_{i,z_i}^{T}\tilde{\theta}_{i,z_i}+\frac{1}{2}\tilde{\theta}_{i,1}^{T}\tilde{\theta}_{i,1}+\frac{1}{2}|\epsilon^{*}|^{2}+|\theta_{i,1}|^{2}.\end{aligned} \tag{69}$$

Step n_i: Proceeding analogously to Step j, we derive $\dot{V}_{i,n_i}$ as follows:

$$s_{i,n_i}=\hat{x}_{i,n_i}-\alpha_{i,n_i-1}-\kappa_i^{(n_i-1)} \tag{70}$$

$$\begin{aligned}\dot{s}_{i,n_i}&=\dot{\hat{x}}_{i,n_i}-\dot{\alpha}_{i,n_i-1}-\kappa_i^{(n_i)}\\&=\varpi^{T}u+\hat{\theta}_{i,n_i}^{T}D_{i,n_i}(\hat{x}_{i,n_i})+l_{i,n_i}\tilde{x}_{i,n_i}-\dot{\alpha}_{i,n_{i-1}}-\kappa_i^{(n_i)}\end{aligned} \tag{71}$$

where

$$\varpi^{T}u=\sum_{j\in\{j_i,j_q\}}\varpi_j\bar{u}_j+\sum_{j\in\{j_i,j_q\}}\tau_j\varpi_j r_j u_0. \tag{72}$$

Then, $\dot{s}_{i,n_i}$ can be expressed as

$$\begin{aligned}\dot{s}_{i,n_i}=&\dot{\hat{x}}_{i,n_i}-\dot{\alpha}_{i,n_i-1}-\kappa_i^{(n_i)}\\=&\sum_{j\in\{j_i,j_q\}}\varpi_j\bar{u}_j+\sum_{j\in\{j_i,j_q\}}\tau_j\varpi_j r_j u_0\\&-l_{i,n_i}\tilde{x}_{i,n_i}+\hat{\theta}_{i,n_i}^{T}D_{i,n_i}(\hat{\bar{x}}_{i,n_i})-\tilde{\theta}_{i,n_i}^{T}D_{i,n_i}(\hat{\bar{x}}_{i,n_i})\\&+\tilde{\theta}_{i,n_i}^{T}D_{i,n_i}(\hat{\bar{x}}_{i,n_i})-\Bigg(\frac{\partial\alpha_{i,n_i-1}}{\partial x_{i,1}}(\tilde{x}_{i,2}+\Delta\varphi_{i,1}\\&+\tilde{\theta}_{i,1}^{T}D\left(\hat{x}_{i,1}\right)+\varepsilon_{i,1}\Big)+\sum_{d=1}^{n_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\kappa_i^{(d-1)}}\kappa_i^{(d)}\Bigg)\\&-\kappa_i^{(n_i)}+H_{i,n_i}\end{aligned} \tag{73}$$

where

$$\begin{aligned}H_{i,n_i}=&-\sum_{d=1}^{n_i-1}\frac{\partial\alpha_{i,n_{i-1}}}{\partial\hat{x}_{i,d}}\left(\hat{x}_{i,d+1}+\hat{\theta}_{i,d}^{T}D_{i,d}(\hat{\bar{x}}_{i,d})+l_{i,d}\tilde{x}_{i,1}\right)\\&-\frac{\partial\alpha_{i,n_{i-1}}}{\partial x_{i,1}}\left(\hat{x}_{i,2}+\hat{\theta}_{i,1}^{T}(\hat{x}_{i,1})\right)-\sum_{d=1}^{n_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\hat{\theta}_{i,d}}\dot{\hat{\theta}}_{i,d}.\end{aligned} \tag{74}$$

Consider the following Lyapunov function:

$$V_{i,n_i}=V_{i,n_i-1}+\frac{1}{2}s_{i,n_i}^{2}+\frac{1}{2\gamma_{i,n_i}}\tilde{\theta}_{i,n_i}^{T}\tilde{\theta}_{i,n_i}. \tag{75}$$

We can get the differentiation of V_{i,z_i} as

$$\begin{aligned}\dot{V}_{i,n_i}&=\dot{V}_{i,n_i-1}+s_{i,n_i}\dot{s}_{i,n_i}-\frac{1}{\gamma_{i,n_i}}\tilde{\theta}_{i,n_i}^{T}\dot{\hat{\theta}}_{i,n_i}\\&=\dot{V}_{i,n_i-1}+s_{i,n_i}\Big(\varpi^{T}u+\hat{\theta}_{i,n_i}^{T}D(\hat{\bar{x}}_{i,n_i})+l_{i,n_i}\tilde{x}_{i,n_i}\\&\quad-\sum_{d=1}^{n_i-1}\frac{\partial\alpha_{i,d-1}}{\partial\kappa_i^{(d-1)}}\kappa_i^{(d)}+\tilde{\theta}_{i,n_i}^{T}D(\hat{\bar{x}}_{i,n_i})-\tilde{\theta}_{i,n_i}^{T}D(\hat{\bar{x}}_{i,n_i})\end{aligned}$$

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$$- \frac{\partial \alpha_{i,z_i-1}}{\partial x_{i,1}} \left(\tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}_{i,1}^T D(\hat{x}_{i,1}) + \epsilon_{i,1} \right) - H_{i,n_i} - \kappa_i^{(n_i)} \Big) - \frac{1}{\gamma_{i,n_i}} \tilde{\theta}_{i,n_i} \dot{\hat{\theta}}_{i,n_i}. \tag{76}$$

Using Young's inequality, we arrive at the following inequalities:

$$\begin{aligned} & - s_{i,n_i} \frac{\partial \alpha_{i,n_i-1}}{\partial x_{i,1}} \left(\tilde{x}_{i,2} + \Delta\varphi_{i,1} + \tilde{\theta}_{i,1}^T D(\hat{x}_{i,1}) + \epsilon_{i,1} \right) \\ \leq \; & 2 s_{i,n_i}^2 \left(\frac{\partial \alpha_{i,n_i-1}}{\partial x_{i,1}} \right)^2 + \frac{1}{2} \epsilon_{i,1}^{*2} + \frac{1}{2} \left| \tilde{x}_i \right|^2 + \left| \theta_{i,1} \right|^2 \\ & + \frac{1}{2} \tilde{\theta}_{i,1}^T \tilde{\theta}_{i,1}, \end{aligned} \tag{77}$$

$$- s_{i,n_i} \tilde{\theta}_{i,n_i}^T D(\hat{\tilde{x}}_{i,n_i}) \leq \pi s_{i,n_i}^2 + \frac{1}{\pi} \tilde{\theta}_{i,n_i}^T \tilde{\theta}_{i,n_i}. \tag{78}$$

The control input u_0 is constructed based on the fuzzy blended rules as

$$\begin{aligned} \sum_{j=j_1 \cdots j_q} \tau_j \varpi_j r_j u_0 = & - \sum_{j=j_1 \cdots j_q} \varpi_j \bar{u}_j - \hat{\theta}_{i,n_i}^T D_{i,n_i}(\hat{\tilde{x}}_{i,n_i}) \\ & - l_{i,n_i} \tilde{x}_{i,n_i} + \kappa_i^{(n_i)} + \sum_{d=1}^{n_i-1} \frac{\partial \alpha_{i,d-1}}{\partial \kappa_i^{(d-1)}} \kappa_i^{(d)} \\ & - \bar{H}_{i,n_i} - c_{i,n_i} s_{i,n_i} - s_{i,n_i-1} - \pi s_{i,n_i}. \end{aligned} \tag{79}$$

By rearranging the aforementioned equation, the explicit expression of the controller can be written as

$$\begin{aligned} u_0 = (g')^{-1} \Bigg(& - \sum_{j=j_1 \cdots j_q} \varpi_j \bar{u}_j - \hat{\theta}_{i,n_i}^T D_{i,n_i}(\hat{\tilde{x}}_{i,n_i}) + \kappa_i^{(n_i)} \\ & - l_{i,n_i} \tilde{x}_{i,n_i} + \sum_{d=1}^{n_i-1} \frac{\partial \alpha_{i,d-1}}{\partial \kappa_i^{(d-1)}} \kappa_i^{(d)} - \bar{H}_{i,n_i} - c_{i,n_i} s_{i,n_i} \\ & - \pi s_{i,n_i} - s_{i,n_i-1} \Bigg) \end{aligned} \tag{80}$$

and $\dot{\hat{\theta}}_{i,n_i}$ is constructed as

$$\dot{\hat{\theta}}_{i,n_i} = \Xi_{i,n_i} D_{i,n_i}(\hat{\tilde{x}}_{i,n_i}) s_{i,n_i} - \rho_{i,n_i} \hat{\theta}_{i,n_i} \tag{81}$$

where c_{i,n_i} and ρ_{i,n_i} are positive constants and $g' = \sum_{j \neq j_1 \ldots j_q} \tau_j \varpi_j r_j$.

By substituting (76)–(81) into V_{i,n_i}, we obtain

$$\begin{aligned} \dot{V}_{i,n_i} \leq & - \sum_{d=1}^{n_i} \left(c_{i,d} s_{i,d}^2 + \frac{\rho_{i,d}}{2\gamma_{i,d}} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} \right) + \frac{n_i}{2} \left| \tilde{x}_{i,1} \right|^2 \\ & + \sum_{d=2}^{n_i} \frac{1}{\pi} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} + \frac{n_i - 1}{2} \tilde{\theta}_{i,1}^T \tilde{\theta}_{i,1} + \frac{n_i \left| \epsilon^* \right|^2}{2} \\ & + \sum_{d=1}^{n_i} \frac{\rho_{i,d}}{2\gamma_{i,d}} \theta_{i,d}^T \theta_{i,d} + n_i \left| \theta_{i,1} \right|^2. \end{aligned} \tag{82}$$

Finally, we have

$$\begin{aligned} \dot{V}_i \leq & - a_{xi} |\tilde{x}_i|^2 + |P_i|^2 \sum_{d=1}^{n_i} \frac{1}{2} \tilde{\omega}_{i,d}^T \tilde{\omega}_{i,d} + \frac{n_i}{2} \left| \tilde{x}_{i,1} \right|^2 + n_i \left| \theta_{i,1} \right|^2 \\ & - \sum_{d=1}^{n_i} \left(c_{i,d} s_{i,d}^2 + \frac{\rho_{i,d}}{2\gamma_{i,d}} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} \right) + \sum_{d=2}^{n_i} \frac{1}{\pi} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} \\ & + \frac{n_i - 1}{2} \tilde{\theta}_{i,1}^T \tilde{\theta}_{i,1} + \tau_i \end{aligned} \tag{83}$$

where $\tau_i = \frac{n_i |\epsilon^*|^2}{2} + \sum_{d=1}^{n_i} \frac{\rho_{i,d}}{2\gamma_{i,d}} \theta_{i,d}^T \theta_{i,d} + \tau_{xi}$.

We define

$$\begin{aligned} C_1 = \min \Bigg\{ & \frac{2\lambda_n}{\lambda_{\max}(P)}, \; 2c_{i,d}, \sigma_1 - \gamma_1 \left(|P|^2 + n - 1 \right), \\ & \sigma_i - \gamma_i \left(|P|^2 + \frac{2}{\pi} \right), \; i = 2, 3, \ldots, n \Bigg\}. \end{aligned}$$

Based on the derivation of Lyapunov analysis and (83), the inequality $\dot{V} \leq -C_1 V + \bar{D}$ guarantees that the property of semiglobal uniform ultimate boundedness holds for all signals within the closed-loop system.

F. Excluding Zeno Behavior

We now show that the proposed DETMs do not exhibit Zeno behavior. Since two event-triggered sequences are introduced in the present framework, namely $\{t_i^k\}_{k \in \mathbb{N}}$ for the transmission of v_i and $\{\varsigma_i^k\}_{k \in \mathbb{N}}$ for the transmission of p_i, it suffices to prove that both sequences admit strictly positive lower bounds on their interevent times.

1) Step 1—Exclusion of Zeno Behavior for the Sequence $\{t_i^k\}_{k \in \mathbb{N}}$: Fix any agent $i \in \{1, \ldots, N\}$. Suppose, by contradiction, that the event sequence $\{t_i^k\}_{k \in \mathbb{N}}$ has a finite accumulation point $T_{v,i}^* < \infty$, namely

$$\lim_{k \to \infty} t_i^k = T_{v,i}^*. \tag{84}$$

For $t \in [t_i^k, t_i^{k+1})$, define the event-triggered error as

$$e_{v,i}(t) = \breve{v}_i(t) - v_i(t) \tag{85}$$

$$\breve{v}_i(t) = v_i(t_i^k). \tag{86}$$

Since $\breve{v}_i(t)$ is constant on $[t_i^k, t_i^{k+1})$, one has

$$\dot{e}_{v,i}(t) = -\dot{v}_i(t) \tag{87}$$

for almost all $t \in [t_i^k, t_i^{k+1})$.

Moreover, by the distributed optimization dynamics in (19)

$$\dot{v}_i(t) = \alpha_4 \sum_{j=1}^{N} a_{ij} \left(\breve{p}_i(t) - \breve{p}_j(t) \right) \tag{88}$$

where $\breve{p}_j(t)$ denotes the latest transmitted value of p_j available at time t, i.e.,

$$\breve{p}_j(t) = p_j(\varsigma_j^{k_j'(t)}) \tag{89}$$

$$\varsigma_j^{k_j'(t)} = \max\{\varsigma_j^k : \varsigma_j^k \leq t\}. \tag{90}$$

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From the boundedness analysis established in the previous subsections, all closed-loop signals are bounded on any compact time interval. Hence, on $[0, T^*_{v,i}]$, there exists a positive constant $\bar{P}_{v,i}$ such that

$$|\breve{p}_i(t)| \le \bar{P}_{v,i} \tag{91}$$

$$|\breve{p}_j(t)| \le \bar{P}_{v,i} \qquad \forall j \in \mathcal{N}_i \tag{92}$$

for all $t \in [0, T^*_{v,i}]$. Therefore,

$$|\dot{e}_{v,i}(t)| = |\dot{v}_i(t)| \le \alpha_4 \sum_{j=1}^{N} a_{ij} \left(|\breve{p}_i(t)| + |\breve{p}_j(t)|\right)$$

$$\le 2\alpha_4 \bar{P}_{v,i} \sum_{j=1}^{N} a_{ij} =: \bar{\Phi}_{v,i} \tag{93}$$

for almost all $t \in [t_i^k, t_i^{k+1})$. Thus, $e_{v,i}(t)$ is absolutely continuous on every interevent interval and satisfies

$$|e_{v,i}(t_i^{k+1})| = \left| \int_{t_i^k}^{t_i^{k+1}} \dot{e}_{v,i}(s)\, ds \right| \le \int_{t_i^k}^{t_i^{k+1}} |\dot{e}_{v,i}(s)|\, ds$$

$$\le \bar{\Phi}_{v,i} \left(t_i^{k+1} - t_i^k\right). \tag{94}$$

On the other hand, by the triggering rule (20), one has

$$\phi_{v,i}(t_i^{k+1}) \ge 0. \tag{95}$$

Using the definition of $\phi_{v,i}$ in (22), it follows that

$$a_v |e_{v,i}(t_i^{k+1})|^2 \ge b_v \left| \sum_{j=1}^{N} a_{ij} \left(\breve{v}_j(t_i^{k+1}) - \breve{v}_i(t_i^{k+1})\right) \right|^2$$

$$+ h_{v,i} \varrho_{v,i}(t_i^{k+1}) \ge h_{v,i} \varrho_{v,i}(t_i^{k+1}). \tag{96}$$

Hence,

$$|e_{v,i}(t_i^{k+1})| \ge \sqrt{\frac{h_{v,i}}{a_v}\, \varrho_{v,i}(t_i^{k+1})}. \tag{97}$$

Moreover, from (30)–(31), the auxiliary variable $\varrho_{v,i}(t)$ remains strictly positive and satisfies

$$\varrho_{v,i}(t) \ge e^{-(f_{v,i}+h_{v,i})t} \varrho_{v,i}(0) \qquad \forall t \ge 0. \tag{98}$$

Since $t_i^{k+1} \le T^*_{v,i}$, we have

$$\varrho_{v,i}(t_i^{k+1}) \ge e^{-(f_{v,i}+h_{v,i})T^*_{v,i}} \varrho_{v,i}(0) =: \underline{\varrho}_{v,i} > 0. \tag{99}$$

Substituting the aforementioned estimate into (97) yields

$$|e_{v,i}(t_i^{k+1})| \ge \sqrt{\frac{h_{v,i}}{a_v}\, \underline{\varrho}_{v,i}} =: \underline{e}_{v,i} > 0. \tag{100}$$

If $\bar{\Phi}_{v,i} > 0$, then combining (94) and (100) leads to

$$t_i^{k+1} - t_i^k \ge \frac{\underline{e}_{v,i}}{\bar{\Phi}_{v,i}} =: \tau_{v,i}^{\min} > 0. \tag{101}$$

If $\bar{\Phi}_{v,i} = 0$, then $\dot{e}_{v,i}(t) \equiv 0$ for almost all $t \in [t_i^k, t_i^{k+1})$. Since $e_{v,i}(t_i^k) = 0$, it follows that

$$e_{v,i}(t) \equiv 0 \qquad \forall t \in [t_i^k, t_i^{k+1}). \tag{102}$$

Consequently,

$$\phi_{v,i}(t) = -h_{v,i} \varrho_{v,i}(t) - b_v \left| \sum_{j=1}^{N} a_{ij} \left(\breve{v}_j(t) - \breve{v}_i(t)\right) \right|^2 < 0 \tag{103}$$

because $\varrho_{v,i}(t) > 0$. Hence, no new triggering instant can occur on this interval, and Zeno behavior is impossible.

Therefore, in all cases, the sequence $\{t_i^k\}_{k\in\mathbb{N}}$ admits a strictly positive lower bound on its interevent times, which contradicts the assumption that $t_i^k \to T^*_{v,i} < \infty$. Hence, Zeno behavior is excluded for the sequence $\{t_i^k\}_{k\in\mathbb{N}}$.

2) Step 2—Exclusion of Zeno Behavior for the Sequence $\{\varsigma_i^k\}_{k\in\mathbb{N}}$*:* Now consider the event sequence $\{\varsigma_i^k\}_{k\in\mathbb{N}}$. Suppose, by contradiction, that it has a finite accumulation point $T^*_{p,i} < \infty$, namely

$$\lim_{k\to\infty} \varsigma_i^k = T^*_{p,i}. \tag{104}$$

For $t \in [\varsigma_i^k, \varsigma_i^{k+1})$, define

$$e_{p,i}(t) = \breve{p}_i(t) - p_i(t) \tag{105}$$

$$\breve{p}_i(t) = p_i(\varsigma_i^k). \tag{106}$$

Since $\breve{p}_i(t)$ is constant on $[\varsigma_i^k, \varsigma_i^{k+1})$, one has

$$\dot{e}_{p,i}(t) = -\dot{p}_i(t) \tag{107}$$

for almost all $t \in [\varsigma_i^k, \varsigma_i^{k+1})$.

By (18)

$$\dot{p}_i(t) = -\alpha_1 \nabla g_i(y_i(t)) - \alpha_2 \sum_{j=1}^{N} a_{ij} \left(\breve{p}_i(t) - \breve{p}_j(t)\right)$$

$$- \alpha_3 \sum_{j=1}^{N} a_{ij} \left(\breve{v}_i(t) - \breve{v}_j(t)\right) \tag{108}$$

where $\breve{v}_j(t)$ denotes the latest transmitted value of v_j available at time t, i.e.,

$$\breve{v}_j(t) = v_j(t_j^{k'_j(t)})$$

$$t_j^{k'_j(t)} = \max\{t_j^k : t_j^k \le t\}. \tag{109}$$

Again, from the boundedness analysis established in the previous subsections, the signals y_i, p_i, and v_i are bounded on $[0, T^*_{p,i}]$. Since $\nabla g_i(\cdot)$ is Lipschitz continuous by Assumption 1, it is bounded on bounded sets. Hence, there exist positive constants $\bar{G}_i$, $\bar{P}_{p,i}$, and $\bar{V}_{p,i}$ such that

$$|\nabla g_i(y_i(t))| \le \bar{G}_i \tag{110}$$

$$|\breve{p}_i(t)| \le \bar{P}_{p,i} \tag{111}$$

$$|\breve{p}_j(t)| \le \bar{P}_{p,i} \qquad \forall j \in \mathcal{N}_i \tag{112}$$

$$|\breve{v}_i(t)| \le \bar{V}_{p,i} \tag{113}$$

$$|\breve{v}_j(t)| \le \bar{V}_{p,i} \qquad \forall j \in \mathcal{N}_i \tag{114}$$

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for all $t \in [0, T^*_{p,i}]$. Therefore,

$$|\dot{e}_{p,i}(t)| = |\dot{p}_i(t)| \leq \alpha_1 \bar{G}_i + 2\alpha_2 \bar{P}_{p,i} \sum_{j=1}^{N} a_{ij} + 2\alpha_3 \bar{V}_{p,i} \sum_{j=1}^{N} a_{ij} =: \bar{\Phi}_{p,i} \tag{115}$$

for almost all $t \in [\varsigma_i^k, \varsigma_i^{k+1})$. Hence,

$$|e_{p,i}(\varsigma_i^{k+1})| = \left| \int_{\varsigma_i^k}^{\varsigma_i^{k+1}} \dot{e}_{p,i}(s)\, ds \right| \leq \int_{\varsigma_i^k}^{\varsigma_i^{k+1}} |\dot{e}_{p,i}(s)|\, ds \leq \bar{\Phi}_{p,i} \left(\varsigma_i^{k+1} - \varsigma_i^k \right). \tag{116}$$

At the triggering instant ς_i^{k+1}, by (21) and (22),

$$\phi_{p,i}(\varsigma_i^{k+1}) \geq 0 \tag{117}$$

which implies

$$a_p |e_{p,i}(\varsigma_i^{k+1})|^2 \geq b_p \left| \sum_{j=1}^{N} a_{ij} \left(\breve{p}_j(\varsigma_i^{k+1}) - \breve{p}_i(\varsigma_i^{k+1}) \right) \right|^2 + h_{p,i} \varrho_{p,i}(\varsigma_i^{k+1}) \geq h_{p,i} \varrho_{p,i}(\varsigma_i^{k+1}). \tag{118}$$

Thus,

$$|e_{p,i}(\varsigma_i^{k+1})| \geq \sqrt{\frac{h_{p,i}}{a_p} \varrho_{p,i}(\varsigma_i^{k+1})}. \tag{119}$$

Using (30)–(31), one further has

$$\varrho_{p,i}(t) \geq e^{-(f_{p,i}+h_{p,i})t} \varrho_{p,i}(0) \qquad \forall t \geq 0. \tag{120}$$

Since $\varsigma_i^{k+1} \leq T^*_{p,i}$, it follows that

$$\varrho_{p,i}(\varsigma_i^{k+1}) \geq e^{-(f_{p,i}+h_{p,i})T^*_{p,i}} \varrho_{p,i}(0) =: \underline{\varrho}_{p,i} > 0. \tag{121}$$

Substituting the aforementioned inequality into (119) gives

$$|e_{p,i}(\varsigma_i^{k+1})| \geq \sqrt{\frac{h_{p,i}}{a_p} \underline{\varrho}_{p,i}} =: \underline{e}_{p,i} > 0. \tag{122}$$

If $\bar{\Phi}_{p,i} > 0$, then combining (116) and (122) yields

$$\varsigma_i^{k+1} - \varsigma_i^k \geq \frac{\underline{e}_{p,i}}{\bar{\Phi}_{p,i}} =: \tau^{\min}_{p,i} > 0. \tag{123}$$

If $\bar{\Phi}_{p,i} = 0$, then $\dot{e}_{p,i}(t) \equiv 0$ for almost all $t \in [\varsigma_i^k, \varsigma_i^{k+1})$. Since $e_{p,i}(\varsigma_i^k) = 0$, it follows that

$$e_{p,i}(t) \equiv 0 \qquad \forall t \in [\varsigma_i^k, \varsigma_i^{k+1}). \tag{124}$$

Consequently,

$$\phi_{p,i}(t) = -h_{p,i} \varrho_{p,i}(t) - b_p \left| \sum_{j=1}^{N} a_{ij} \left(\breve{p}_j(t) - \breve{p}_i(t) \right) \right|^2 < 0 \tag{125}$$

because $\varrho_{p,i}(t) > 0$. Hence, no new triggering instant can occur on this interval, and Zeno behavior is impossible.

Therefore, in all cases, the sequence $\{\varsigma_i^k\}_{k\in\mathbb{N}}$ admits a strictly positive lower bound on its interevent times, which contradicts the assumption that $\varsigma_i^k \to T^*_{p,i} < \infty$.

Hence, both event-triggered sequences $\{t_i^k\}_{k\in\mathbb{N}}$ and $\{\varsigma_i^k\}_{k\in\mathbb{N}}$ are free of Zeno behavior. The proof is completed.

Theorem 1: A class of nonlinear MASs (2) is investigated, operating in accordance with Assumptions 1–6. Based on the distributed optimization DETC algorithms (18), (19), a resilient distributed observer (46)–(48) is designed. Together with the adaptive laws (57), (68), (81) and the designed observer (9) with actuator failure compensation control (80), the proposed adaptive distributed optimization strategy guarantees that:

1) under the proposed actuator-fault-tolerant controller, all signals in the resulting closed-loop system are bounded despite the presence of actuator failures;
2) the discrepancy between the output of each agent y_i and the global minimizer y^* converges to a small neighborhood of zero;
3) the observer can provide bounded estimates of the unknown states;
4) both event-triggered sequences $\{t_i^k\}_{k\in\mathbb{N}}$ and $\{\varsigma_i^k\}_{k\in\mathbb{N}}$ are free of Zeno behavior.

Proof: The Lyapunov function is selected as

$$V = \sum_{i=1}^{N} V_i + \delta V_\xi \tag{126}$$

where δ is a positive design constant.

From Assumption 1, one has $\tilde{p}^\top \tilde{\nabla}_g(p, p^*) > 0$. Combining the observer error analysis, the backstepping recursive design, and the Lyapunov analysis of the $(\tilde{p}, \tilde{v})$-subsystem yields

$$\begin{aligned} \dot{V} \leqslant & -a_i |\tilde{x}_i|^2 + |P_i|^2 \sum_{d=1}^{n_i} \frac{1}{2} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} - \delta K^T \tilde{Q} K - \delta \alpha_7 |\tilde{v}|^2 \\ [2mm] & - \bar{c}_{i,1} s_{i,1}^2 - \sum_{d=2}^{n_i} c_{i,d} s_{i,d}^2 - \sum_{d=1}^{n_i} \frac{\rho_{i,d}}{2\gamma_{i,d}} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} \\ & + \sum_{d=2}^{n_i} \frac{1}{\pi} \tilde{\theta}_{i,d}^T \tilde{\theta}_{i,d} + \frac{n_i - 1}{2} \tilde{\theta}_{i,1}^T \tilde{\theta}_{i,1} + \tau_i \end{aligned} \tag{127}$$

where the coefficients are explicitly defined as

$$a_i = a_{x_i} - \frac{n_i}{2} > 0, \qquad \bar{c}_{i,1} = c_{i,1} - \delta \alpha_8 > 0. \tag{128}$$

By appropriately choosing the design parameters such that all coefficients of the quadratic terms remain positive, there exist two positive constants $C > 0$ and $\bar{D} > 0$ satisfying

$$\dot{V} \leq -CV + \bar{D}. \tag{129}$$

Therefore, the signals $\tilde{x}_i$, $s_{i,\xi}$, $\tilde{\theta}_{i,d}$, $\tilde{p}_i$, and $\tilde{v}_i$ are all bounded. Since $\tilde{p}_i = p_i - y^*$, it follows that p_i is bounded. Consequently, the virtual control signals α_i and the system states x_i are bounded. Moreover, since $\hat{x}_i = x_i - \tilde{x}_i$, the observer states $\hat{x}_i$

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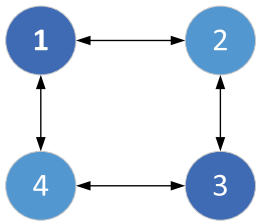


Fig. 2. Communication topology.

are also bounded. This proves that all signals in the closed-loop system remain bounded and that the observer can provide bounded estimates of the unknown states.

Furthermore, it follows that there exists a bounded vicinity ε_i such that

$$\limsup_{t\to\infty}|y_i(t)-y^*|\le\limsup_{t\to\infty}|s_{i,1}(t)|+\limsup_{t\to\infty}|\tilde{p}_i(t)|\le\varepsilon_i. \tag{130}$$

That is, the output tracking error converges to a small neighborhood of zero, which demonstrates the effectiveness of the proposed control strategy in ensuring approximate tracking performance.

IV. Simulation Studies

This section presents a simulation case to validate the proposed methodology. To demonstrate the effectiveness of the proposed event-triggered distributed adaptive controller, we consider $N=4$ interconnected mass–spring–damper subsystems, each described by

$$\begin{aligned}\dot{x}_{i,1}&=x_{i,2}+\varphi_{i,1}(x_{i,1},x_{i,2}),\\ \dot{x}_{i,2}&=\frac{-1}{\bar{m}_i}\left(\bar{k}_i x_{i,1}+\zeta_i x_{i,2}-\varphi_{i,2}(x_{i,1},x_{i,2})\right)\\ &\quad+\varpi_{i,1}u_{i,1}+\varpi_{i,2}u_{i,2}.\end{aligned} \tag{131}$$

where $x_{i,1}$ and $x_{i,2}$ denote the displacement and velocity, respectively, and $\bar{m}_i$, ζ_i, $\bar{k}_i$ are the mass, damping, and stiffness parameters. The nonlinear perturbations are modeled as

$$\varphi_{i,1}(x_{i,1},x_{i,2})=\bar{\alpha}_1 x_{i,1}\sin(\beta_1 x_{i,1})+\bar{\alpha}_2 x_{i,2}^2\cos(\beta_2 x_{i,1}) \tag{132}$$

$$\varphi_{i,2}(x_{i,1},x_{i,2})=\bar{\gamma}_1 x_{i,2}\sin(\gamma_2 x_{i,1})+\bar{\gamma}_3 x_{i,1}^3+\bar{\gamma}_4 x_{i,1}x_{i,2}. \tag{133}$$

With their communication topology illustrated in Fig. 2. The associated Laplacian matrix is defined as $\mathcal{L}=\begin{bmatrix}2&-1&0&-1\\-1&2&-1&0\\0&-1&2&-1\\-1&0&-1&2\end{bmatrix}$.

The initial values and parameters are set as $\alpha_1=0.28$, $\alpha_2=0.16$, $\alpha_3=0.16$, and $\alpha_4=0.15$, $x_{1,1}(0)=-0.1$, $x_{1,2}(0)=-0.1, x_{2,1}(0)=-0.2, x_{2,2}(0)=-0.1, x_{3,1}(0)=-0.2, x_{3,2}(0)=-0.2, x_{4,1}(0)=-0.1, x_{4,2}(0)=-0.1$, $[\hat{x}_{i1}(0),\hat{x}_{i2}(0)]=[-0.1,-0.3]$, $i=1,2,3,4$. The observer gains are chosen as $l_{i1}=l_{i2}=3.5$ for all agents. The optimization variables are initialized by $p_i(0)=[-0.1,-0.2,-0.1,-0.1]$ and $v_i(0)=[-0.1,-0.5,-0.1,-0.1]$.

TABLE I
Simulation Parameters for Each Agent

Parameter	Agent 1	Agent 2	Agent 3	Agent 4
$\bar{m}_i$	1.0	1.2	0.9	1.1
$\bar{k}_i$	2.0	2.1	2.2	2.0
ζ_i	0.15	0.10	0.20	0.12
$\bar{\alpha}_{1,2}$	0.4, 0.2	0.3, 0.2	0.4, 0.25	0.35, 0.2
$\bar{\beta}_{1,2}$	1.0, 1.2	0.9, 1.0	1.1, 1.3	1.0, 1.1
$\bar{\gamma}_{1\text{–}4}$	(0.3, 0.2, 0.1, 0.05)	same	same	same
$y^{\text{ref}}_{1\text{–}4}$	1	same	same	same

Note: The entry "same" indicates that the parameter value is identical to that of Agent 1.

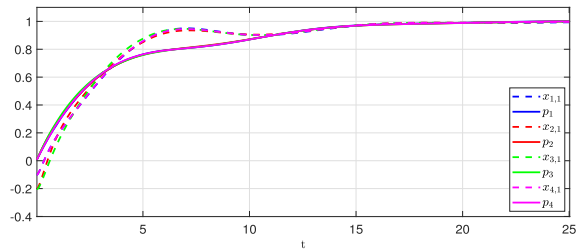


Fig. 3. Tracking performance.

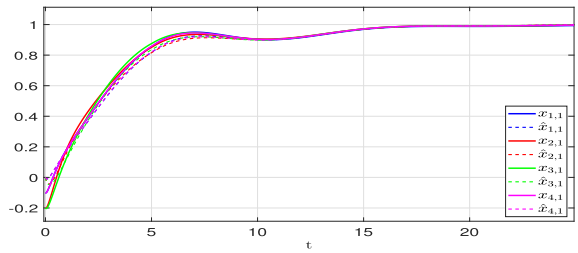


Fig. 4. State and observation.

Each agent i aims to minimize a local convex cost function $\Lambda_i(y_i)=\frac{1}{2}(y_i-y_i^{\text{ref}})^2$, where gradient $\nabla\Lambda_i(y_i)=y_i-y_i^{\text{ref}}$ is employed to generate the reference signal p_i via the distributed optimization (18)–(19). Table I lists the physical and controller parameters used in the simulation.

Fig. 2 shows the output trajectories $y_i(t)$ of all agents and the optimal reference $y^*(t)$, demonstrating that each subsystem successfully tracks the global optimum under actuator degradation and nonlinear perturbations. Fig. 3 presents the output trajectories $y_i(t)$ and the optimal reference y^*. Fig. 4 shows the system states and their observations. Fig. 5 depicts the interevent intervals under the proposed DETMs. Fig. 6 shows the control inputs. Fig. 7 presents the filtered reference states $\zeta_{i,1}$.

The results verify that the proposed adaptive fuzzy backstepping controller with actuator fault compensation ensures robust tracking performance and optimization convergence even in the presence of nonlinear perturbations and limited communication resources.

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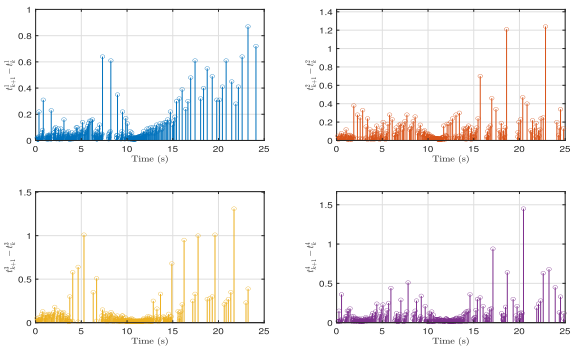


Fig. 5. Interevent intervals $t_{k+1}^i - t_k^i$ under DETMs.

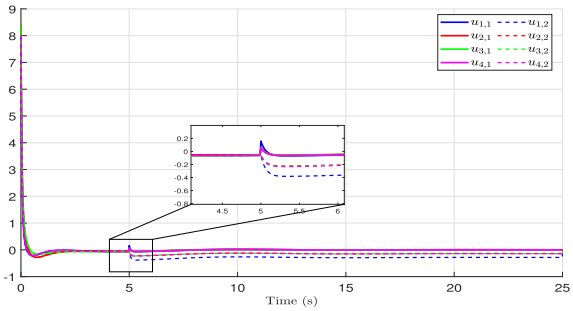


Fig. 6. Trajectories of control.

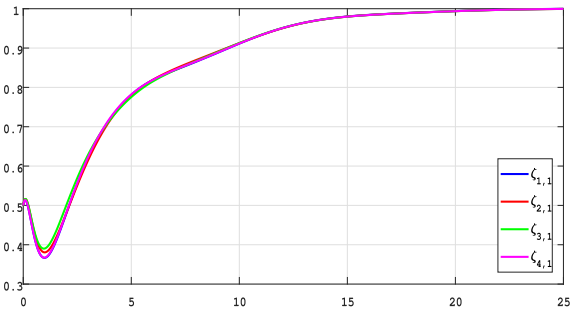


Fig. 7. Trajectory of $\zeta_{i,1}$.

V. Conclusion

In this study, a distributed adaptive optimization architecture was constructed to address uncertain nonlinear MASs under a DETM. A unified output-feedback fault-tolerant control framework was developed by integrating a distributed optimization law, a high-order smoothing filter, a fuzzy state observer, and an adaptive backstepping controller to cope with unknown dynamics and actuator faults. Theoretical analysis guaranteed the boundedness of all signals in the closed-loop system and the convergence of tracking errors to a small vicinity of the origin. The validity and robustness of the suggested control strategy were demonstrated through simulation studies. The present results were established for fixed undirected communication graphs, strongly convex objective functions, and delay-free information exchange. Hence, extensions to directed or switching topologies, communication delays, and nonconvex optimization settings remain open. Practical implementation issues such as online computational burden, measurement quality, and real-time triggering evaluation will also be investigated in future work.

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